



Logistic Regression in Clinical Studies

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Highlights

- A logistic regression model is used when the outcome of interest is binary. The term “logistic” refers to the underlying “logit” (log odds) function that is used to model the binary outcome.
- Odds ratios are produced from a logistic regression model and have a useful interpretation.
- Tips, tricks and concepts used to fit logistic regression models are similar to those used in linear regression models.
- Modeling building that is knowledge-based rather than automatic is preferred in most applications of logistic regression.
- A logistic regression model that is overparameterized (ie, too many variables for too few events) can result in odds ratios that are implausibly large and confidence intervals that are wide and uninterpretable. These types of “overfitted” models should be avoided.
- Logistic regression models can be fit using most standard statistical software. © 2021 Elsevier Inc. All rights reserved.

What Is Logistic Regression?

Suppose a clinician is interested in examining whether a new method of delivering radiation therapy reduces the onset of a specific toxicity after taking into account patient demographics and clinicopathologic features. In another very different example, suppose lightbulb engineers want to investigate what environmental conditions are likely to affect the odds that a new brand of lightbulb will fail in the first year of use. These are 2 examples in which a logistic regression model, the topic of this paper, could be useful. In the case vignette, we will take you through many of the steps involved in fitting a logistic regression model.

A logistic regression model describes and estimates the relationship between 1 binary dependent variable, also known as an outcome variable, and 1 or more independent variables, also known as covariates or explanatory variables. Logistic regression models are versatile, have a powerful interpretation, and have been used to describe phenomena in diverse areas of medical and nonmedical research. Similar to other regression models, a logistic regression model is often used to evaluate predictors and to adjust for confounders and/or interactions. These models are used to analyze retrospective data, including case-control studies, as well as to create prediction algorithms, which can be depicted in nomograms or online calculators that communicate the probability of an event, eg toxicity or

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lightbulb failure in the 2 previously noted examples. Logistic regression models are also used to calculate propensity scores that can then be used to balance case mix (ie, many covariates) of individuals in 2 cohorts when the goal is to compare the 2 cohorts on an outcome. The propensity scores created from a logistic regression model are often used in a matched or weighted analysis.¹ Finally, because the outcome variable is binary, logistic regression can be used in 2-class classification problems, and as a result, it is a commonly used machine-learning algorithm.

The model

Linear regression models are relatively straightforward to intuitively understand, as the covariates are placed into a function or equation that predicts a continuous outcome that can theoretically range from negative to positive infinity; for any given patient in the data set, insert their covariate values into the equation, and the model will estimate this patient's continuous outcome.

However, the linear regression model used to describe a continuous outcome cannot be used to model an outcome that has just 2 possible results (eg, 0 or 1, yes or no, heads or tails). If we were to do this, we would get a terrible fit to the data. The elegant solution of logistic regression converts the binary outcome into a continuous one, the log odds, or logit, of that event.²⁻⁴ Indeed, this model is just one in the category of generalized linear models, which also includes the linear regression model and the Poisson regression model, among others. In logistic regression, rather than directly modeling the binary outcome, the log-odds, also called the logit function, is modeled. In the context of logistic regression, this equation is given algebraically as:

$$\begin{aligned} \text{logit}(p_x) &= \log\left(\frac{p_x}{1-p_x}\right) \\ &= \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k \end{aligned} \quad (1)$$

In this equation, the intercept is given by β_0 , and $\beta_1 \dots \beta_k$ are the parameter estimates associated with the covariates $X_1 \dots X_k$. The intercept term, β_0 , is the log of the odds of an event (ie, $Y = 1$) when all covariates are set to zero. The β_k s have an extremely useful and important interpretation, which we will describe and demonstrate in later sections. Note that in Equation 1 you may recognize the logit function $p/(1-p)$ as the "odds." The odds of an event occurring is defined as the probability that the event occurs divided by the probability that the event does not occur. The probability of an event occurring and its complement, the probability of the event not occurring, must sum to 1. When the odds are greater than 1, the probability of the event happening is higher than the probability of the event not happening, and when the odds are less than 1, the probability of the event happening is less than the probability of the event not happening.

With further algebraic manipulation of Equation 1, one can see how a binary variable, Y , fits into this equation. Solving for p_x , Equation 1 can be rearranged as follows:

$$P(Y = 1 | X) = \left(\frac{\exp^{\beta_0 + \beta_1 X + \cdots + \beta_k X_k}}{1 + \exp^{\beta_0 + \beta_1 X + \cdots + \beta_k X_k}} \right) \quad (2)$$

Equation 2 is useful because it shows how the binary outcome Y is converted to probabilities. For each patient in the data set, one can calculate the probability that the patient has the event of interest from the the natural exponent $[\exp(x)]$ of the sum of the product of each of the covariates $X_1 \dots X_k$ and their corresponding parameter estimates. As seen in the vignette, consider acute toxicity (yes vs no) as the outcome. If there is 1 continuous covariate, X , in the model, the probability of having an acute toxicity can be calculated for a patient as $\left(\frac{\exp^{\beta_0 + \beta_1 X}}{1 + \exp^{\beta_0 + \beta_1 X}} \right)$. One can think of these probabilities as fitted or predicted values that range from 0 to 1, in which a probability of 1 indicates that a patient (with fixed covariate values) has a 100% probability of having the event.

As a result of this formulation, the regression parameters produced by logistic regression, $\beta_1 \dots \beta_k$ in the previously noted equations, represent log-odds ratios. The odds ratio is the ratio of 2 odds. The log odds ratio is the natural logarithm of the odds ratio. Therefore, the coefficients indicate the amount of change expected in the log odds when there is a 1-unit change in the predictor variable with all of the other variables in the model held constant.

As a sidenote, the odds ratio is not the risk ratio or relative risk, which is based on the ratio of 2 probabilities. In certain cases (eg, when the prevalence of the outcome of interest is very low), the risk ratio can be approximated from the odds ratio.⁵

Interpretation of the odds ratio

Once a logistic model is fit, model coefficients can be exponentiated to give the odds ratio (OR). An OR equal to 1 means that the odds of the event in 1 group are the same as the odds of the event in another group; there is no difference. An OR >1 means that 1 group has a higher odds of having the event compared with the reference group. Finally, an OR <1 means that 1 group has a lower odds of having an event compared with the reference group. More detailed examples on how to interpret odds ratios are given in the Case Study section.

In addition, the standard error of each coefficient (and odds ratio) can be used to construct a confidence interval and perform hypothesis tests. The Wald test is used to evaluate whether the coefficient (ie, the log odds ratio) is significantly different from zero or, equivalently, whether the odds ratio is different from 1 (indicating no effect).

When Can I Use Logistic Regression?

As with most statistical tests and models, there are assumptions underlying logistic regression. These assumptions are similar to those underlying linear regression.

1. Units of analysis are sampled independently (i.i.d.; *identically, independently distributed*)
2. The model fit is correct. This means that for any continuous independent variables, there should be a linear relationship between a continuous variable and the respective logit-transformed outcomes (see [Equation 2](#)). For example, given a continuous variable V , the odds ratio for V changing from 10 to 20 should be similar to the odds ratio for V changing from 50 to 60.
3. Independent variables are not collinear. A logistic regression model with highly correlated independent variables can result in large standard errors for the odds ratio estimates.
4. Lack of strongly influential outliers.

There are additional characteristics of the data set to consider that will affect the ability to fit a logistic regression model. First, does the data set have enough events to even do a logistic regression analysis? A model that is overparameterized (“overfit”) could result in odds ratios that are implausibly large and confidence intervals that are wide and uninterpretable,^{6,7} as the model hinges on just a few distinct outcomes. Logistic regression models should include only the number of covariates, or nonreference variable levels in the case of categorical variables, in which the 10 events rule-of-thumb is met. This means that if you have a data set of 1000 observations but only 5 events, a logistic regression model will not fit well and should be avoided as the primary analysis, as at least 10 events are needed to support inclusion of even a single continuous or binary covariate. If there were 20 events in a data set of 1000 observations, then a maximum of 2 covariates can be considered in a multivariable logistic regression model. This rule is demonstrated in practice in the Case Study section.

The feasibility and validity of fitting a logistic regression model are also directly affected by missing values present in any of the covariates considered. Although there are sophisticated methods to manage missing data, in general, if an observation is missing a value on variable X_k , then this observation is automatically excluded from any model that includes X_k .⁸ Coupled with the 10 events per variable rule, the power to fit a logistic regression model can be reduced significantly if missingness occurs with high frequency in some or many covariates. Thus, in addition to ensuring the 4 assumptions are met, it is important for the investigator to do a preliminary check on the degree of missingness and number of events before fitting any logistic regression model to a data set.

Case Study

For patients with recurrence or second primary squamous cell carcinoma in the head and neck after treatment with radiation therapy, some may be offered reirradiation using intensity modulated radiation therapy (IMRT). To investigate factors associated with acute toxicity among reirradiated patients in this setting, a multicenter retrospective analysis was conducted among 505 patients who were initially treated with ≥ 40 Gy of radiation to the head and/or neck and then underwent re-IMRT to ≥ 40 Gy after development of recurrent or second primary squamous cell carcinoma. The setting, methods, and results of the original study have been reported in detail previously.⁹

In this case vignette, we will examine a subset of the variables included in univariable logistic regression analyses for the outcome of acute toxicity, and will explore various options for building a multivariable logistic regression model. We will then interpret the results, though our goal is not to draw any new clinical conclusions from these data, but rather simply to demonstrate an application of logistic regression in a radiation oncology setting. Finally, we will examine model fit using various diagnostics and discuss common issues to look out for when conducting a logistic regression analysis.

The binary outcome of interest in this case vignette is acute grade 3 toxicity, specifically including admission to the hospital for aspiration pneumonia, new tracheostomy use, new feeding tube placement, esophageal stricture dilation, neutropenic fever, or soft tissue necrosis. If a patient experienced any of these acute grade 3 toxicities, they will be coded as a “yes” with the value 1 for the outcome whereas if a patient did not experience any of these acute toxicities they will be coded as a “no” with the value of 0 for the outcome. Included are 462 patients who have complete data on acute toxicity available; 102 (22.1%) patients experienced an acute toxicity. This means that we can include up to 10 variables, or nonreference levels of categorical variables, in a multivariable logistic regression model while still adhering to the 10 events per variable rule.

Univariable logistic regression

A common first step in building a multivariable logistic regression model is to examine univariable logistic regression results for each of the factors of interest. “Univariable” refers to a model with 1 explanatory, or independent, variable included. In this case study, we will examine whether age, sex, Charlson comorbidity score, Karnofsky performance status, rT stage, rN stage, systemic chemotherapy, and re-IMRT dose are associated with acute toxicity. The univariable logistic regression results are presented in [Table 1](#). We find that only rT stage is significantly associated with acute toxicity ($P = .002$) at the 0.05 significance level, such that patients with rT4 disease have 2.02 times the odds (ie, an odds ratio of 2.02) of acute toxicity (95% confidence interval [CI], 1.29-3.17) compared with patients

Table 1 Univariable logistic regression results

Characteristic	N	OR	95% CI	P value
Age (per 1 year)	462	1.01	0.99, 1.03	.5
Sex	462			.2
Male		—	—	
Female		1.37	0.84, 2.21	
Charlson comorbidity score	459			.6
0		—	—	
1		1.30	0.74, 2.23	
≥2		1.25	0.71, 2.14	
Karnofsky performance status	422			.5
>70		—	—	
≤70		1.23	0.64, 2.29	
rT stage	462			.002
All others		—	—	
rT4		2.02	1.29, 3.17	
rN stage	409			.4
N0		—	—	
N+		1.24	0.78, 1.97	
Systemic chemotherapy	461			.052
No		—	—	
Yes		1.75	0.99, 3.23	
Re-IMRT dose	462			.4
<66		—	—	
≥66		0.81	0.50, 1.30	

Abbreviations: CI = confidence interval; OR = odds ratio.

with all other rT stages of disease. We also note that there are missing data for some variables, specifically Charlson comorbidity score (3 missing), Karnofsky performance status (40 missing), rN stage (53 missing), and systemic chemotherapy (1 missing). Missing values will be excluded from each univariable logistic regression model, where applicable. It is important to consider missing values when building a multivariable logistic regression model, as missing values for any variable included in a multivariable model will be excluded from the model, and the sample size can be affected importantly in some cases.

These univariable models include continuous, categorical, and binary predictors. Some variable types have additional considerations. For categorical variables, we should consider the hypotheses being tested. If you are not explicitly testing a separate hypothesis for each level of a categorical variable compared with a reference level, then you should present a

single, overall *P* value. Table 2 shows the univariable logistic regression model results for Charlson comorbidity score with level-specific *P* values on the left and an overall *P* value on the right. Level-specific *P* values would be appropriate if you wished to test separate hypotheses regarding the associations between Charlson comorbidity score 1 versus 0 and Charlson comorbidity scores ≥2 versus 0 with respect to acute toxicity. An overall *P* value would be appropriate if you wished to test the single hypothesis of whether Charlson comorbidity score is associated with acute toxicity. For continuous variables, we may wish to consider the variable scale. In Table 1 we saw results for a 1-year increase in age. But often with a small per-unit change such as this, we will have an odds ratio and confidence interval very close to 1, which can be difficult to interpret. We could rather choose to present results for age for a 2-year, 5-year, 10-year, or other relevant increase. Table 3 compares the univariable logistic regression results for age per a 1-year increase and age per a 10-year increase. We see that although the *P* values remain exactly the same, the odds ratio for acute toxicity is 1.07 (95% CI, 0.87-1.31) for a 10-year increase in age compared with 1.01 (95% CI, 0.99-1.03) for a 1-year increase in age. Implicit in this presentation is that increased odds of toxicity for someone 50 years old versus 40 years old is the same as the increased odds for someone 70 years old versus 60 years old.

Multivariable logistic regression

One of the most important steps in a logistic regression analysis is selecting which variables to include in a multivariable logistic regression model. “Multivariable” refers to a model with more than 1 explanatory variable (eg, independent variable or covariate). In logistic regression, multivariable models are important to adjust for covariate imbalances that may exist by outcome and to control for confounders that are commonly found in nonrandomized studies. A *confounder* refers to a variable that is associated with both the outcome and an independent variable of interest. When a confounder is omitted from a model, resulting model estimates will be biased. For example, a study may find high coffee consumption to be associated with increased lung cancer incidence. In this example, smoking status is a likely confounder because high coffee consumers may also have a heavier smoking history. One can imagine that the odds ratio for coffee consumption could be highly attenuated once

Table 2 Comparison of level-specific versus overall *P* values for categorical variables in logistic regression

Characteristic	Level-specific <i>P</i> values				Overall <i>P</i> value			
	N	OR	95% CI	<i>P</i> value	N	OR	95% CI	<i>P</i> value
Charlson comorbidity score	459				459			.6
0		—	—			—	—	
1		1.30	0.74, 2.23	0.4		1.30	0.74, 2.23	
≥2		1.25	0.71, 2.14	0.4		1.25	0.71, 2.14	

Abbreviations: CI = confidence interval; OR = odds ratio.

Table 3 Comparison of univariable logistic regression results for age per a 1-year increase or per a 10-year increase

Characteristic	N	OR	95% CI	P value
Age (per 1 year)	462	1.01	0.99, 1.03	.5
Age (per 10 years)	462	1.07	0.87, 1.31	.5

Abbreviations: CI = confidence interval; OR = odds ratio.

smoking was also included in the model. Therefore, it is important to account for all possible confounding variables.¹⁰

Identifying confounders is easiest when there is 1 predictor or independent variable of primary interest, for example, treatment A versus treatment B. In this paradigm, all other patient and disease characteristics can be considered as potential confounders, and one could evaluate the association between the primary predictor (eg, treatment A vs B) and the binary outcome after including each potential confounder. If the association changes substantially, this is a hint that confounding may exist and additional investigation may be necessary.

With a number of potential covariates and confounders in hand, how does one effectively build a multivariable model in a systematic, thoughtful, and robust way? Just as with multivariable linear regression, there are a number of ways to select variables for inclusion in a multivariable model, and here we will discuss the pros and cons of each and make recommendations about how to proceed. Commonly used variable selection procedures can be broken down into 3 main categories. First are variable selection procedures such as forward, backward, and stepwise selection, which select variables to include based on significance either on their own or in combination with other variables. Although these procedures are easy to implement with many available statistical software programs, they are generally not recommended. These approaches do not take into consideration the clinical importance of variables that may affect the model even if not statistically significant. For example, it is often common in clinical research to include age and sex in multivariable regression models regardless of statistical significance because we know these are important patient characteristics that often influence outcomes. Additionally, selecting variables simply based on statistical significance can lead to undesirable statistical properties of the final model such as overfit and questionable interpretation of the final *P* values. This occurs primarily because automatic variable selection procedures conduct many repeated statistical tests, thereby creating a multiple testing scenario in which the pre-established overall type I error rate (usually 0.05) no longer holds and consequently the risk of finding a significant covariate by chance is higher. In other words, regression coefficients from automated variable selection procedures are often inflated (larger estimated effects), and their confidence intervals are overly narrow, thus giving the false impression of larger effect sizes with higher precision. Conducting the analysis in separate training and test data sets is 1 way to use these automated procedures while lessening the consequences of multiple testing.

A second approach is to select variables based only on clinical importance, such as variables known to influence the

outcome or to be confounders based on prior literature or variables that are of specific interest based on a clinical hypothesis of association with the outcome. This method is an ideal approach in the context of clinical research, because our fundamental goal is to explain associations of variables of interest with outcomes. However, the drawback of this approach is that although there are often a large number of relevant clinical variables that are suited for inclusion, a modest number of observed outcomes may limit the appropriate number of acceptable covariates that can be used in a multivariable model, given the importance to adhere to the 10 events per variable rule. So we may still need a way to select a subset of the variables that are of interest clinically.

The final approach is the one that we recommend when all variables of clinical interest cannot be included in the model owing to sample size limitations, which is to use a combination of clinical and statistical significance. In this approach we would start with all predictors that are of clinical interest, and then we can use statistical significance along with goodness-of-fit statistics to compare among different models that include an appropriately sized subset of these clinically important predictors.

Returning to our case study, we see that the variables considered in univariable logistic regression represent a total of 9 variables or nonreference variable levels: each of the 7 continuous and binary variables represents 1 variable and the 3-level categorical variable Charlson comorbidity score represents 2 nonreference variable levels. Note that a multivariable model containing all of these covariates would be based on 367 patients, rather than the full patient sample of 462, as patients missing any of the included predictors were excluded from the model. As a result of these exclusions for missingness, there are 83 acute toxicity events observed in our sample. This means that we cannot include all of these variables in the multivariable model based on the 10 events per variable rule, so we will need to select 1 variable to exclude. Based on the clinical interest here, we will exclude sex from the model as there is no specific hypothesis that sex would influence acute toxicity in this setting. We find that with age, Charlson comorbidity score, Karnofsky performance status, rT stage, rN stage, systemic chemotherapy, and re-IMRT dose included in the model, at the 0.05 level rT stage is significantly associated with acute toxicity, such that patients with rT4 disease have 2.08 times the odds (ie, an odds ratio of 2.08) of acute toxicity (95% CI, 1.21-3.56) compared with patients with all other rT stages of disease and holding all other variables constant. The addition of systemic chemotherapy also significantly increased the odds of acute toxicity (ie, odds ratio of 1.94; 95% CI, 1.01-3.95) (Table 4). All other covariates had *P* values > .05, so they would not be considered statistically significant at the 5% type I error level.

Variable selection

Finally, we consider models that include a subset of predictors included in the full multivariable model and use Akaike's

Table 4 Full multivariable logistic regression model

Characteristic	OR	95% CI	P value
Age (per 1 year)	1.01	0.98, 1.04	.4
Charlson comorbidity score			.6
0	—	—	
1	1.36	0.69, 2.61	
≥2	1.30	0.67, 2.46	
Karnofsky performance status			.4
>70	—	—	
≤70	1.37	0.67, 2.70	
rT stage			.008
All others	—	—	
rT4	2.08	1.21, 3.56	
rN stage			.2
N0	—	—	
N+	1.40	0.83, 2.38	
Systemic chemotherapy			.046
No	—	—	
Yes	1.94	1.01, 3.95	
Re-IMRT dose			.4
<66	—	—	
≥66	0.80	0.46, 1.36	

Abbreviations: CI = confidence interval; IMRT = intensity modulated radiation therapy; OR = odds ratio.

information criterion (AIC) to compare different models according to model fit. When the number of events supports it, inclusion of all variables that are of a priori clinical interest is the ideal approach to model building. However, we will not always have a sufficient number of events to support all of the variables that are of a priori clinical interest and so at times will need to compare different models and decide what subset of variables to include. The AIC is a measure of prediction error, so smaller values indicate lower prediction error and therefore a comparatively better model.

One common approach is to split variables into subsets based on type and compare models that include different subsets. For example, we may split our variables into 3 subsets: patient characteristics (ie, age, sex, Charlson comorbidity score, Karnofsky performance status), disease characteristics (ie, rT stage, rN stage), and treatment characteristics (ie, systemic chemotherapy, re-IMRT dose). We may be sure that we want to include the patient characteristics, but then we want to choose among models that additionally contain the disease characteristics or the treatment characteristics. The resulting multivariable model with age, sex, Charlson comorbidity score, Karnofsky performance status, rT stage, and rN stage has an AIC of 396.5, whereas the multivariable model with age, sex, Charlson comorbidity score, Karnofsky performance status, systemic chemotherapy, and re-IMRT dose has an AIC of 446.4. So in this case we may choose the model including disease characteristics versus the model including treatment characteristics because it has the lower AIC.

At times we may be even more limited in terms of the number of variables our sample size can support. In this case study we are specifically interested in variables associated with acute toxicity. So from a clinical perspective the

Table 5 AIC for multivariable logistic regression models adding each variable in the table to a model already containing age, sex, systemic chemotherapy, and re-IMRT dose

Added variable	AIC
Charlson comorbidity score	489.1
Karnofsky performance status	447.0
rT stage	483.5
rN stage	442.5

Abbreviations: AIC = Akaike's information criteria; IMRT = intensity modulated radiation therapy.

most important variables may be systemic chemotherapy and re-IMRT dose, as we hypothesize that receipt of systemic chemotherapy and higher re-IMRT dose may both be associated with increased odds of acute toxicity. We may also know that it is important to include age and sex to account for heterogeneity of patient characteristics. So we will start with a base model that includes age, sex, systemic chemotherapy, and re-IMRT dose. In an example in which we are only able to include 1 additional variable, we can consider models that add each additional variable one at a time and compare them based on the AIC. Here we may select the model with age, sex, systemic chemotherapy, re-IMRT dose, and rN stage as it has the lowest AIC (Table 5).

One additional consideration during the variable selection process is collinearity. Collinearity, or multicollinearity, results when 1 or more predictors in a model are highly related. Collinearity can result in unstable parameter estimates in the model which can lead to uncertain interpretation of the results, and erroneous conclusions about significance or lack of significance of individual predictors. If 2 variables usually point in the same direction, the model will strongly depend on just a handful of discrepant patients, which will lead to unstable estimates. Some signs of collinearity include large standard errors that result in wide confidence intervals, for the affected variables or large changes in parameter estimates of the affected variables when there are small changes in the data. But clinical expertise can also be used to detect collinearity. For example, in our case study we have considered both age and Charlson comorbidity score in our multivariable model at times. But we know that age is 1 of the inputs used to calculate some versions of the Charlson comorbidity score, so by design these 2 variables can have some degree of positive correlation that may lead to collinearity in the model. We could consider models leaving 1 or the other of these 2 variables out and assess how the estimates and standard errors for the other variables change as a result of each omission.

Interaction effects

Lastly, at times it may be of interest to investigate if there is effect modification in the multivariable logistic regression model by generating interaction terms. Effect modification is the scenario in which the effect of 1 independent variable on the outcome changes depending on levels of another

independent variable. For example, in the original publication by Caudell et al (2017), a significant interaction was identified between elective neck irradiation (unilateral or bilateral vs none) and the treatment intent (postoperative vs definitive) with respect to acute grade ≥ 3 toxicity, such that in the definitive setting, elective neck irradiation was not associated with acute grade ≥ 3 toxicity but in the postoperative setting, elective neck irradiation was significantly associated with increased odds of acute grade ≥ 3 toxicity. In other words, the effect of elective neck irradiation on acute grade ≥ 3 toxicity differed depending on the treatment setting. Operationally, an interaction effect is tested by including the 2 independent variables as well as the interaction as a covariate. In the example, the interaction is defined as the product defined by elective neck irradiation $\times$ treatment intent, where each variable is coded as 0 or 1. The P value of the interaction term indicates whether the difference in the effect estimates is statistically significant. In general, interaction effects require larger sample sizes and should be investigated only when there is a strong underlying scientific hypothesis regarding a possible interaction.

Other Forms of Logistic Regression

As mentioned, logistic regression requires independent identically distributed (iid) observations. However, in some studies, cases are matched to controls thereby violating this assumption when case-control status is the binary outcome. In this circumstance, a conditional logistic regression model that accounts for the matched study design could be fit to the data. A matched study design in this case is 1 in which cases (eg, rare cancer patients) are matched 1:1 or 1:k to controls (controls that do not have cancer) on certain predefined patient or disease characteristics. To evaluate whether covariates (those not used in the matching process) are associated with being a case, a conditional logistic regression model that adjusts for the matching process can be fit.

In this paper we considered binary outcomes. Ordinal or categorical outcomes with more than 2 levels, known as polytomous outcomes, can also be modeled using techniques related to logistic regression. In an ordinal regression model, the outcome is assumed to be categorical but with a natural ordering (eg, 0 lymph nodes involved, 2-4 nodes involved, >4 nodes involved or complete response, partial response, stable, progressive). In a multinomial regression model, the outcome is assumed to be categorical with no natural ordering (eg, blood types A, B, AB, O). Both of these models produce odds ratios, as in logistic regression.

Conclusion

Logistic regression models are commonly used to assess the relationship between 1 or more independent variables with a binary outcome. Variable selection is critical and uses the same processes as methods used for other regression

techniques. Logistic regression is powerful yet significant errors can be made in model building that can mar the validity and interpretation of the results. Consequently, and as is true for any model building exercise, it is important to evaluate whether underlying model assumptions are met, and these evaluations and other decisions made during model building must be carefully explained in the final report.

Dos and Don'ts

- Do look at the degree of missingness and distributions of each covariate.
- Do look at the odds ratios and confidence intervals to ensure both are scientifically plausible.
- Do not overparameterize. A useful rule of thumb is to include only the number of covariates in which the 10 events per variable rule is met.
- Do evaluate model fit and competing models using the AIC.
- Ensure that the relationship between continuous covariate and the outcome is linear and before model building.
- Do consider possible confounders when building a multivariable model.
- Do test for interaction effects when there is a specific hypothesis to back it up.

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